



3.1

Addressing Modes (8)

3.1.1 Addressing Modes: NIELIT 2016 DEC Scientist B (CS) - Section B: 53



The addressing mode used in an instruction of the form $ADD\ X\ Y$, is

- A. Direct B. Absolute C. Indirect D. Indexed

nielit2016dec-scientistb-cs co-and-architecture addressing-modes

Answer key

3.1.2 Addressing Modes: NIELIT 2016 DEC Scientist B (IT) - Section B: 55



The example of implied addressing is

- A. Stack addressing B. Indirect addressing
C. Immediate addressing D. None of the above

nielit2016dec-scientistb-it co-and-architecture addressing-modes

Answer key

3.1.3 Addressing Modes: NIELIT 2016 MAR Scientist B - Section C: 9



A certain processor supports only the immediate and the direct addressing modes. Which of the following programming language features cannot be implemented on this processor?

- A. Pointers. B. Arrays. C. Records. D. All of these.

nielit2016mar-scientistb co-and-architecture addressing-modes

Answer key

3.1.4 Addressing Modes: NIELIT 2017 DEC Scientist B - Section B: 47



INCA(Increase register A by 1) is an example of which of the following addressing mode?

- A. Immediate addressing B. Indirect addressing
C. Implied addressing D. Relative addressing

nielit2017dec-scientistb co-and-architecture addressing-modes

Answer key

3.1.5 Addressing Modes: NIELIT 2017 DEC Scientist B - Section B: 59



A two-word instruction is stored in a location A . The operand part of instruction holds B . If the addressing mode is relative, the operand is available in location

- A. $A + B + 2$ B. $A + B + 1$ C. $B + 1$ D. $A + B$

nielit2017dec-scientistb co-and-architecture addressing-modes

Answer key 

3.1.6 Addressing Modes: NIELIT 2018-82



Identify the true statement from the given statements. Program relocation at run time

1. requires transfer complete block to some memory locations
2. requires both base address and relative address
3. requires only absolute address

A. 1 B. 1 and 2 C. 1 , 2 and 3 D. 1 and 3

nielit-2018 co-and-architecture addressing-modes

Answer key 

3.1.7 Addressing Modes: NIELIT 2021 Dec Scientist A - Section B: 48



The addressing mode/s, which uses the PC instead of a general-purpose register is :

- | | |
|------------------------|--|
| A. Indexed with offset | B. Relative |
| C. Direct | D. Both Indexed with offset and direct |

nielit2021dec-scientista co-and-architecture addressing-modes

Answer key 

3.1.8 Addressing Modes: NIELIT Scientific Assistant A 2020 November: 117



In the following addressing mode, which of them performs better for accessing, array?

- | | |
|---------------------------------|---------------------------|
| A. Register addressing mode | B. Direct addressing mode |
| C. Displacement addressing mode | D. Index addressing mode |

nielit-sta-2020 co-and-architecture addressing-modes easy

Answer key 

3.2 CO and Architecture (9)

3.2.1 CO and Architecture: NIELIT 2016 DEC Scientist B (CS) - Section B: 55



Process that periodically checks status of an I/O devices, is known as

- A. Cold swapping B. I/O instructions C. Polling D. Dealing

nielit2016dec-scientistb-cs co-and-architecture

Answer key 

3.2.2 CO and Architecture: NIELIT 2016 MAR Scientist C - Section C: 19



When a subroutine is called, then address of the instruction following the CAL instruction is stored in/on the

- A. Stack pointer B. Accumulator C. Program counter D. Stack

nielit2016mar-scientistc co-and-architecture

Answer key 

3.2.3 CO and Architecture: NIELIT 2016 MAR Scientist C - Section C: 27



The process of entering data into a storage location

- A. causes variation in its address number B. adds to the contents of the location
C. is called a readout operation D. is destructive of previous contents

nielit2016mar-scientistc co-and-architecture

Answer key 

3.2.4 CO and Architecture: NIELIT 2016 MAR Scientist C - Section C: 28



Serial access memories are useful in applications where

- A. data consists of numbers B. short access time is required
C. each stored word is processed differently D. data naturally needs to flow in and out in serial form

nielit2016mar-scientistc co-and-architecture

Answer key 

3.2.5 CO and Architecture: NIELIT 2016 MAR Scientist C - Section C: 47



If a processor does not have any stack pointer register, then

- A. it cannot have subroutine call instruction
B. it can have subroutine call instruction, but no nested subroutine calls
C. nested subroutine calls are possible, but interrupts are not
D. all sequences of subroutine calls and also interrupts are possible

nielit2016mar-scientistc co-and-architecture

Answer key 

3.2.6 CO and Architecture: NIELIT 2017 DEC Scientist B - Section B: 38



Which of the following is **false**?

- A. The smallest and fastest computer imitating brain working is called quantum computer.
B. A computer with a speed of around 100 million instructions per second with the word length of around 64 bits is known as super computer.
C. The term Exa-byte = 1024 Tera Bytes.
D. None of the options.

nielit2017dec-scientistb co-and-architecture

Answer key 

3.2.7 CO and Architecture: NIELIT 2017 July Scientist B (CS) - Section B: 56



The part of machine level instruction, which tells the central processor what has to be done, is

- A. Operation code B. Address C. Locator D. Flip-Flop

nielit2017july-scientistb-cs co-and-architecture

Answer key 

3.2.8 CO and Architecture: NIELIT 2018-60



If the period of a signal is 100 ms. Then its frequency in Hertz is _____

- A. 10 B. 100 C. 1000 D. 10000

nielit-2018 co-and-architecture

Answer key 

3.2.9 CO and Architecture: NIELIT 2018-61



The ALU uses _____ to store intermediate result

- A. Cache B. Registers C. Accumulators D. Stack

nielit-2018 co-and-architecture

Answer key 

3.3

Cache Memory (5)

3.3.1 Cache Memory: NIELIT 2017 DEC Scientist B - Section B: 4



In a cache memory if total number of sets are 's', then the set offset is:

- A. 2^8 B. $\log_2 s$ C. s^2 D. s

nielit2017dec-scientistb co-and-architecture cache-memory

Answer key 

3.3.2 Cache Memory: NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 22



In time division switches if each memory access takes 100 ns and one frame period is $125\text{ }\mu\text{s}$, then the maximum number of lines that can be supported is

- A. 625 lines B. 1250 lines C. 2300 lines D. 318 lines

nielit2017oct-assistanta-it co-and-architecture cache-memory

Answer key 

3.3.3 Cache Memory: NIELIT 2021 Dec Scientist A - Section B: 45



More than one word is put in one cache block to:

- A. exploit the temporal locality of reference in a program
- B. exploit the spatial locality of reference in a program
- C. reduce the miss penalty
- D. none of the option

nielit2021dec-scientista co-and-architecture cache-memory

Answer key

3.3.4 Cache Memory: NIELIT 2021 Dec Scientist B - Section B: 15



The size of the physical address space of a processor is 2^P bytes. The word length is 2^W bytes. The capacity of cache memory is 2^N bytes. The size of each cache block 2^M words. For a K -Way set associative cache memory, the length (in number of bits) of the tag fields is:

- A. $P - N - M - W + \log_2 K$
- B. $P - N - M - W - \log_2 K$
- C. $P - N + \log_2 K$
- D. $P - N - \log_2 K$

nielit-2021-it-dec-scientistb co-and-architecture cache-memory nielit2021dec-scientistb

Answer key

3.3.5 Cache Memory: NIELIT Scientist B 2020 November: 69



A direct mapped cache is of size 32 KB and has block size 32 Bytes. CPU also generates 32 bit address. Number of bits needed for indexing the cache:

- A. 14
- B. 15
- C. 10
- D. 17

nielit-scb-2020 cache-memory co-and-architecture

Answer key

3.4 Data Dependency (2)

3.4.1 Data Dependency: NIELIT 2017 OCT Scientific Assistant A (CS) - Section C: 5



Which of the following is a desirable property of module?

- A. Independency
- B. Low cohesiveness
- C. High Coupling
- D. Multifunctional

nielit2017oct-assistanta-cs co-and-architecture data-dependency

Answer key

3.4.2 Data Dependency: NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 10

Which of the following is a desirable property of module?



- A. Independency
- C. High Coupling

- B. Low cohesiveness
- D. Multifunctional

nielit2017oct-assistanta-it co-and-architecture data-dependency

Answer key

3.5

Disk (5)

3.5.1 Disk: NIELIT 2016 DEC Scientist B (CS) - Section B: 50



Where does the swap reside?

- A. RAM
- B. ROM
- C. DISK
- D. On-chip cache

nielit2016dec-scientistb-cs co-and-architecture disk

Answer key

3.5.2 Disk: NIELIT 2017 DEC Scientific Assistant A - Section B: 23



A 3.5 inch micro floppy high density disk contains the data _____ .

- A. 720 MB
- B. 1.44 MB
- C. 720 KB
- D. 1.44 KB

nielit2017dec-assistanta co-and-architecture disk

Answer key

3.5.3 Disk: NIELIT 2017 July Scientist B (CS) - Section B: 22



What is the average Access Time for a drum rotating at 4000 revolutions per minute?

- A. 2.5 milliseconds
- B. 5.0 milliseconds
- C. 7.5 milliseconds
- D. 4.0 milliseconds

nielit2017july-scientistb-cs co-and-architecture disk

Answer key

3.5.4 Disk: NIELIT 2017 OCT Scientific Assistant A (CS) - Section C: 1



Which level of RAID refers to disk mirroring with block striping?

- A. RAID level 1
- B. RAID level 2
- C. RAID level 0
- D. RAID level 3

nielit2017oct-assistanta-cs co-and-architecture disk

Answer key

3.5.5 Disk: NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 26



Which level of RAID refers to disk mirroring with block striping?

- A. RAID level 1 B. RAID level 2 C. RAID level 0 D. RAID level 3

nielit2017oct-assistanta-it co-and-architecture disk

Answer key 

3.6 Instruction Format (6)

3.6.1 Instruction Format: NIELIT 2017 DEC Scientist B - Section B: 40



Which of the following is/are **not** features of RISC processor?

- i. Large number of addressing modes.
- ii. Uniform instruction set.

- A. (i) Only B. (ii) Only C. Both (i) and (ii) D. None of the options

nielit2017dec-scientistb co-and-architecture addressing-modes instruction-format

Answer key 

3.6.2 Instruction Format: NIELIT 2017 DEC Scientist B - Section B: 49



Which of the following is **false**?

- A. Interrupts which are initiated by an instruction are software interrupts
- B. When a subroutine is called, the address of the instruction following the CALL instruction is stored in the stack pointer
- C. A micro program which is written as 0's and 1's is a binary micro program
- D. None of the options

nielit2017dec-scientistb co-and-architecture interrupts instruction-format

Answer key 

3.6.3 Instruction Format: NIELIT 2017 DEC Scientist B - Section B: 6



A stack organized computer has which of the following instructions?

- A. zero-address B. one-address C. two-address D. three-address

nielit2017dec-scientistb co-and-architecture instruction-format

Answer key 

3.6.4 Instruction Format: NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 3

Match list *I* with List *II* and select the correct answer using the codes given below the lists.



	List I		List II
A.	0-address instruction	i.	$T = TOP(T - 1)$
B.	1-address instruction	ii.	$Y = Y + X$
C.	2-address instruction	iii.	$Y = A - B$
D.	3-address instruction	iv.	$ACC = ACC - X$

Codes:

A. A-i, B-ii, C-iii, D-iv

B. A-iii, B-ii, C-iv, D-i

C. A-ii, B-iii, C-i, D-iv

D. A-i, B-iv, C-ii, D-iii

nielit2017oct-assistanta-cs co-and-architecture instruction-format machine-instruction

Answer key 

3.6.5 Instruction Format: NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 35

In a 10-bit computer instruction format, the size of address field is 3-bits. The computer uses expanding OP code technique and has 4 two-address instructions and 16 one-address instructions. The number of zero address instructions it can support is



A. 256

B. 356

C. 640

D. 756

nielit2017oct-assistanta-it co-and-architecture machine-instruction instruction-format

Answer key 

3.6.6 Instruction Format: NIELIT 2017 OCT Scientific Assistant A (IT) - Section C: 8

Match list I with List II and select the correct answer using the codes given below the lists :



	List I		List II
A.	0-address instruction	i.	$T = TOP(T - 1)$
B.	1-address instruction	ii.	$Y = Y + X$
C.	2-address instruction	iii.	$Y = A - B$
D.	3-address instruction	iv.	$ACC = ACC - X$

Codes:

A. A-i, B-ii, C-iii, D-iv

B. A-iii, B-ii, C-iv, D-i

C. A-ii, B-iii, C-i, D-iv

D. A-i, B-iv, C-ii, D-iii

nielit2017oct-assistanta-it co-and-architecture instruction-format machine-instruction

Answer key 

3.7

Interrupts (1)

3.7.1 Interrupts: NIELIT 2016 DEC Scientist B (CS) - Section B: 39



External Interrupt may not arise because of

- A. illegal or erroneous use of an instruction.
- B. a timing device.
- C. external source.
- D. I/O devices.

nielit2016dec-scientistb-cs co-and-architecture interrupts

Answer key

3.8 Machine Instruction (2)

3.8.1 Machine Instruction: NIELIT 2016 DEC Scientist B (IT) - Section B: 39



MIMD stands for

- A. Multiple Instruction Multiple Data
- B. Multiple Instruction Memory Data
- C. Memory Instruction Multiple data
- D. Multiple Information Memory data

nielit2016dec-scientistb-it co-and-architecture machine-instruction

Answer key

3.8.2 Machine Instruction: NIELIT 2016 MAR Scientist C - Section C: 45



An instruction used to set the carry flag in a computer can be classified as

- A. data transfer
- B. process control
- C. logical
- D. program control

nielit2016mar-scientistc co-and-architecture machine-instruction

Answer key

3.9 Memory Interfacing (8)

3.9.1 Memory Interfacing: NIELIT 2016 DEC Scientist B (CS) - Section B: 58



CPU consists of _____.

- A. ALU and Control Unit
- B. ALU, Control Unit and Monitor
- C. ALU, Control Unit and Hard disk
- D. ALU, Control Unit and Register

nielit2016dec-scientistb-cs co-and-architecture memory-interfacing

Answer key

3.9.2 Memory Interfacing: NIELIT 2016 DEC Scientist B (IT) - Section B: 37



How many address lines are needed to address each memory location in a 2048×4 memory chip?

- A. 10
- B. 11
- C. 8
- D. 12

nielit2016dec-scientistb-it co-and-architecture memory-interfacing

Answer key

3.9.3 Memory Interfacing: NIELIT 2017 DEC Scientist B - Section B: 28



A RAM chip has 7 address lines, 8 data lines and 2 chips select lines. Then the number of memory locations is _____

- A. 2^{12} B. 2^{10} C. 2^{19} D. 2^{13}

nielit2017dec-scientistb co-and-architecture memory-interfacing

Answer key

3.9.4 Memory Interfacing: NIELIT 2017 July Scientist B (CS) - Section B: 27



Which memory is difficult to interface with processor?

- A. Static memory B. Dynamic memory C. ROM D. None of the option

nielit2017july-scientistb-cs co-and-architecture memory-interfacing

Answer key

3.9.5 Memory Interfacing: NIELIT 2017 July Scientist B (CS) - Section B: 28



For a memory system, the cycle time is

- A. Same as the access time. B. Longer than the access time.
C. Shorter than the access time. D. Multiple of the access time.

nielit2017july-scientistb-cs co-and-architecture memory-interfacing

Answer key

3.9.6 Memory Interfacing: NIELIT 2017 July Scientist B (CS) - Section B: 29



In comparison with static RAM memory, the dynamic RAM memory has

- A. Lower bit density and higher power consumption B. Higher bit density and lower power consumption
C. Lower bit density and lower power consumption D. None of the option

nielit2017july-scientistb-cs co-and-architecture memory-interfacing

Answer key

3.9.7 Memory Interfacing: NIELIT 2017 July Scientist B (CS) - Section B: 30



If each address space represents one byte of storage space, how many address lines are needed to access RAM chips arranged in a 4×6 array, where each chip is $8K \times 4$ bits?

- A. 13 B. 14 C. 16 D. 17

nielit2017july-scientistb-cs co-and-architecture memory-interfacing

Answer key

3.9.8 Memory Interfacing: NIELIT Scientific Assistant A 2020 November: 55



A 26—bit address bus has maximum accessible memory capacity of _____

- A. 64 MB B. 16 MB C. 1 GB D. 4 GB

nielit-sta-2020 co-and-architecture memory-interfacing

Answer key 

3.10

Memory Management (3)

3.10.1 Memory Management: NIELIT 2017 OCT Scientific Assistant A (CS) - Section C: 2



Which of the following is not a form of main memory?

- A. Instruction cache B. Instruction register
C. Instruction opcode D. Translation look-aside buffer

nielit2017oct-assistanta-cs co-and-architecture memory-management

Answer key 

3.10.2 Memory Management: NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 5



Which of the following is not a form of main memory?

- A. Instruction cache B. Instruction register
C. Instruction opcode D. Translation look-aside buffer

nielit2017oct-assistanta-it co-and-architecture memory-management

Answer key 

3.10.3 Memory Management: NIELIT 2021 Dec Scientist A - Section B: 110



Consider the following sequence of micro operations:

$MBR \leftarrow PC$

$MAR \leftarrow X$

$PC \leftarrow Y$

$MEMORY \leftarrow MBR$

Which one of the following is possible operation performed by this sequence ?

- A. Instruction Fetch B. Operand Fetch
C. Conditional Branch D. Initiation of interrupt service

nielit2021dec-scientista co-and-architecture memory-management

3.11

Microprogramming (4)

3.11.1 Microprogramming: NIELIT 2016 MAR Scientist C - Section C: 46



Micro program is

- A. the name of source program in micro computers
- B. the set of instructions indicating the primitive operations in a system
- C. primitive form of macros used in assembly language programming
- D. program of very small size

nielit2016mar-scientistc co-and-architecture microprogramming

Answer key

3.11.2 Microprogramming: NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 25



A micro programmed control unit

- A. Is faster than a hardwired unit
- B. Facilitates easy implementation of a new instruction
- C. Is useful when small programs are to be run
- D. All of the above

nielit2017oct-assistanta-cs co-and-architecture control-unit microprogramming

Answer key

3.11.3 Microprogramming: NIELIT 2021 Dec Scientist A - Section B: 50



A microprogrammed control unit :

- A. is faster than hardwired control unit
- B. allows easy implementation of new instructions
- C. is useful when small programs are to be run
- D. none of the options

nielit2021dec-scientista co-and-architecture microprogramming

Answer key

3.11.4 Microprogramming: NIELIT Scientist B 2020 November: 54



Consider a control unit generating the control signals. These control signals are divided into five mutually exclusive groups as shown below:

Groups	G_1	G_1	G_1	G_1	G_1
Control Signals	3	7	10	12	2

How many bits are saved using the Vertical Micro-programmed instead of Horizontal Micro-programmed control unit?

- A. 14
- B. 34
- C. 20
- D. None

nielit-scb-2020 co-and-architecture microprogramming

Answer key

3.12 Pipelining (5)

3.12.1 Pipelining: NIELIT 2017 DEC Scientist B - Section B: 13



Consider a non-pipelined machine with 6 stages; the lengths of each stage are 20ns, 10ns, 30ns, 25ns, 40 ns and 15ns respectively. Suppose for implementing the pipelining the machine adds 5 ns of overhead to each stage for clock skew and set up. What is the speed up factor of the pipelining system (ignoring any hazard impact)?

- A. 7 B. 14 C. 3.11 D. 6.22

nielit2017dec-scientistb co-and-architecture pipelining

Answer key

3.12.2 Pipelining: NIELIT 2017 DEC Scientist B - Section B: 14



We have 10-stage pipeline, where the branch target conditions are resolved at stage 5. How many stalls are there for an incorrectly predicted branch?

- A. 5 B. 6 C. 7 D. 4

nielit2017dec-scientistb co-and-architecture pipelining

Answer key

3.12.3 Pipelining: NIELIT 2017 July Scientist B (CS) - Section B: 23



Comparing the time T_1 taken for a single instruction on a pipelined CPU, with time T_2 taken on a non-pipelined but identical CPU, we can say that _____ ?

- A. $T_1 = T_2$ B. $T_1 > T_2$
C. $T_1 < T_2$ D. T_1 is T_2 plus time taken for one instruction fetch cycle

nielit2017july-scientistb-cs co-and-architecture pipelining

Answer key

3.12.4 Pipelining: NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 18



A pipeline is having speed up factor as 10 and operating with efficiency of 80%. What will be the number of stages in the pipeline?

- A. 10 B. 8 C. 13 D. None

nielit2017oct-assistanta-it co-and-architecture pipelining

Answer key

3.12.5 Pipelining: NIELIT 2022 April Scientist B | Section B | Question: 99



A 5 stage pipelined CPU has the following sequence of stages:

- **IF** – instruction fetch from instruction memory
- **RD** – Instruction decode and register read
- **EX** – Execute: ALU operation for data and address computation
- **MA** – Data memory access – for write access, the register read at RD state is used.

- WB – Register write back

Consider the following sequence of instructions:

- $I_1 : L \ R0, loc1; R0 \leq M[loc1]$
- $I_2 : A \ R0, R0; R0 \leq R0 + R0$
- $I_3 : S \ R2, R0; R2 \leq R2 - R0$

Let each stage take one clock cycle.

What is the number of clock cycles taken to complete the above sequence of instructions starting from the fetch of I_1 ?

- A. 8 B. 10 C. 12 D. 15

nielit2022apr-scientistb co-and-architecture pipelining

3.13 Pushdown Automata (1)

3.13.1 Pushdown Automata: NIELIT 2021 Dec Scientist A - Section B: 111



In the case of, Zero-address instruction method the operands are stored in _____.

- A. Registers B. Accumulators C. Push down stack D. Cache

nielit2021dec-scientista co-and-architecture pushdown-automata

Answer key

3.14 Speedup (1)

3.14.1 Speedup: NIELIT 2016 DEC Scientist B (CS) - Section B: 23



A nonpipeline system taken $50ns$ to process a task. The same task can be processed in a six-segment pipeline with a clock cycle of $10ns$. Determinant the speedup ration of the pipeline for 100 tasks. What is the maximum speedup that can be achieved?

- A. 4.90,5 B. 4.76,5 C. 3.90,5 D. 4.30,5

nielit2016dec-scientistb-cs co-and-architecture pipelining speedup

Answer key

3.15 Virtual Memory (1)

3.15.1 Virtual Memory: NIELIT Scientist B 2020 November: 76



In _____ VMs do not simulate the underlying hardware.

- A. Para Virtualization B. Full Virtualization
C. Hardware-Assisted Virtualization D. Network Virtualization

nielit-scb-2020 operating-system virtual-memory

Answer Keys

3.1.1	D
3.1.6	B
3.2.3	D
3.2.8	A
3.3.4	C
3.5.2	B
3.6.2	D
3.7.1	A
3.9.3	A
3.9.8	A
3.11.2	B
3.12.3	B
3.15.1	A

3.1.2	A
3.1.7	B
3.2.4	D
3.2.9	C
3.3.5	C
3.5.3	C
3.6.3	A
3.8.1	A
3.9.4	B
3.10.1	C
3.11.3	B
3.12.4	C

3.1.3	D
3.1.8	D
3.2.5	X
3.3.1	B
3.4.1	A
3.5.4	A
3.6.4	D
3.8.2	C
3.9.5	B
3.10.2	C
3.11.4	C
3.12.5	A

3.1.4	C
3.2.1	C
3.2.6	C
3.3.2	A
3.4.2	A
3.5.5	A
3.6.5	C
3.9.1	D
3.9.6	B
3.10.3	D
3.12.1	C
3.13.1	C

3.1.5	A
3.2.2	C
3.2.7	A
3.3.3	B
3.5.1	C
3.6.1	A
3.6.6	D
3.9.2	B
3.9.7	D
3.11.1	B
3.12.2	D
3.14.1	B